Trace Donor Identification Report

Analysis Name: mock-fo

Analyst: fo2025

Date: 2025-12-11 12:19

Trace Mode Analysis
This report identifies the most likely trace donor from the pedigree individuals based on Y-STR haplotype matching. The probabilities below have been normalized to sum to 100% across all pedigree members.

Summary

4	Verdachte	99.42%	426.7s
Total Individuals Analyzed	Most Likely Donor	Highest Match Probability	Total Runtime

Trace Donor Match Probabilities (Normalized)

The following table shows the normalized match probability for each individual in the pedigree. Probabilities are ranked from highest to lowest, with the most likely trace donor at the top.

Rank	Individual ID	Match Probability	Visual
1	**Verdachte**	**99.4165%**	
2	M-1478	**0.2919%**	
3	M-8462	**0.2853%**	

Rank	Individual ID	Match Probability	Visual
4	M-2853	**0.0062%**	

Interpretation

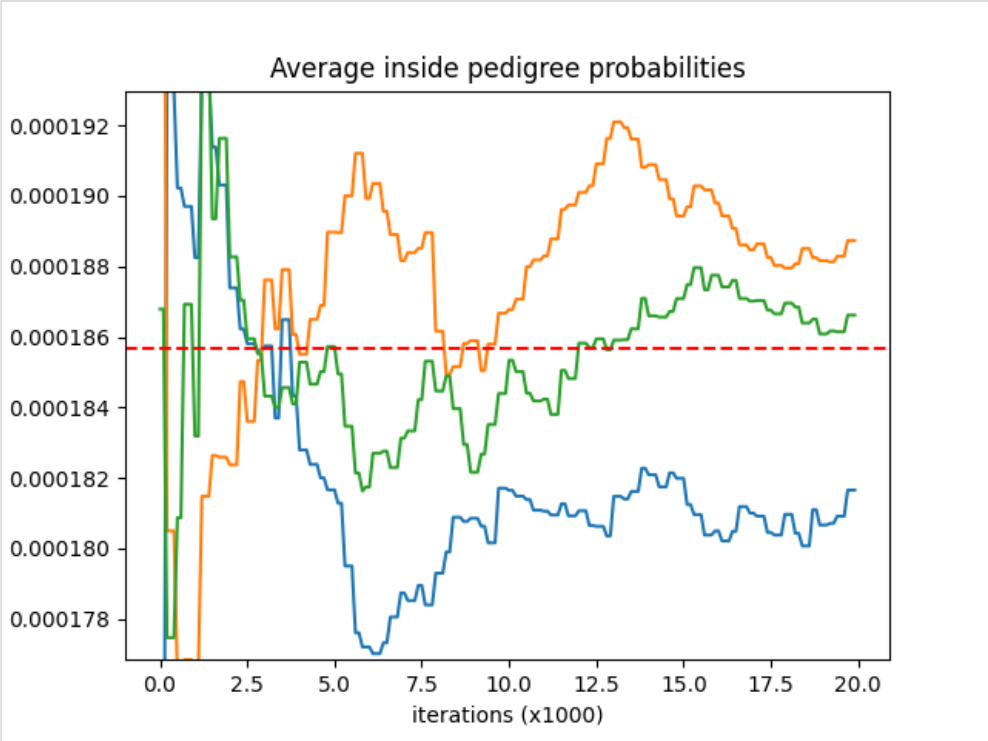
Most Likely Donor: Individual **Verdachte** has the highest normalized match probability of **99.4165%**.

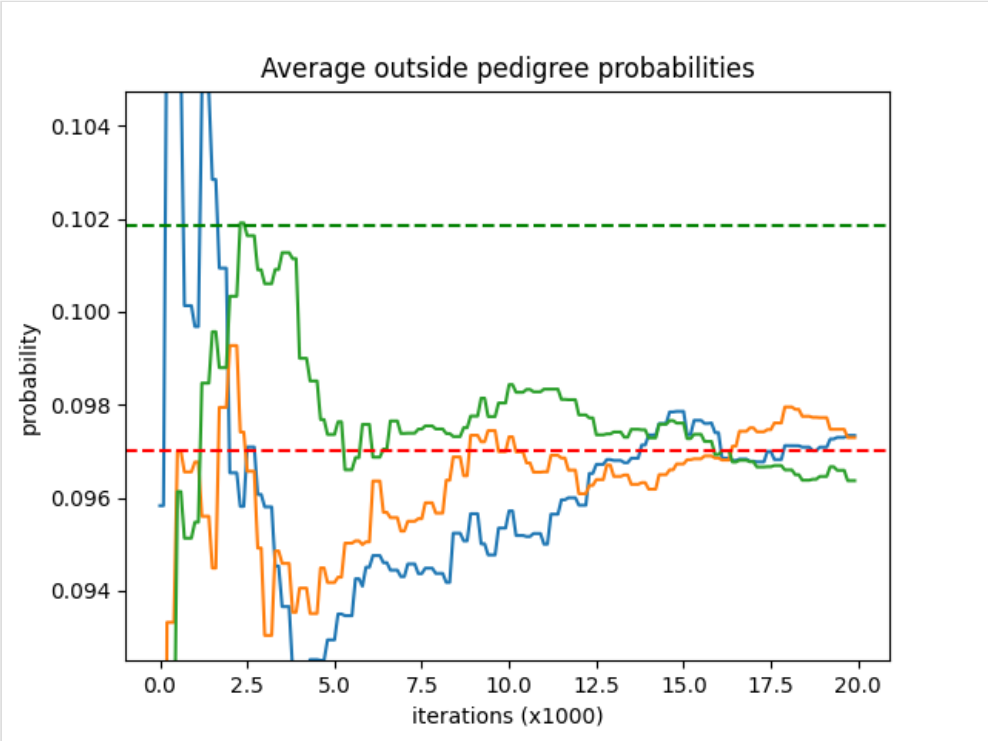
> **Strong Evidence:** The most likely donor has a probability exceeding 50%, suggesting relatively strong evidence that this individual is the trace donor compared to other pedigree members.

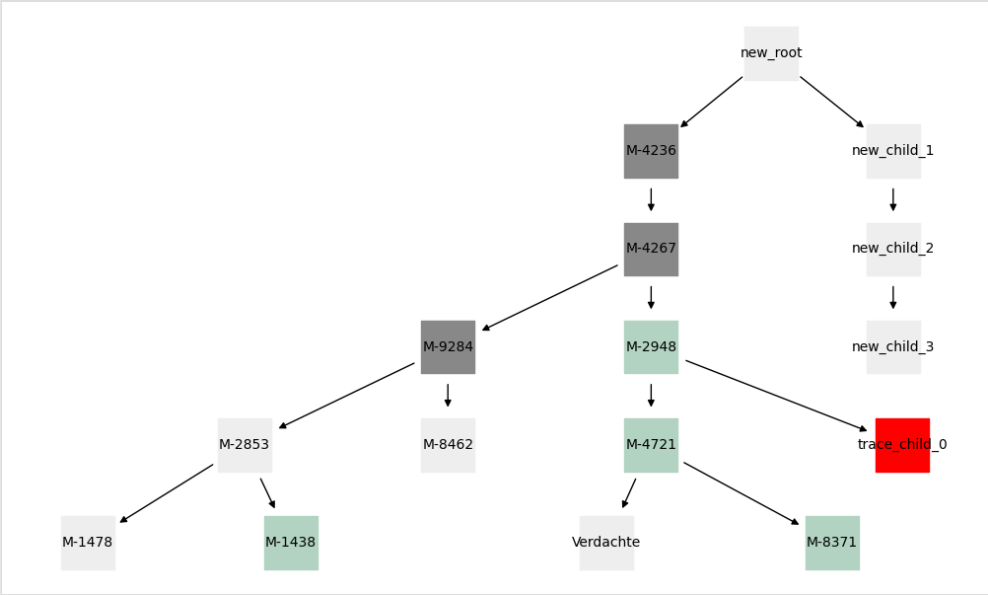
Note: These probabilities represent relative likelihoods within the pedigree and assume the trace donor is among the analyzed individuals. They do not account for the possibility of an unrelated individual.

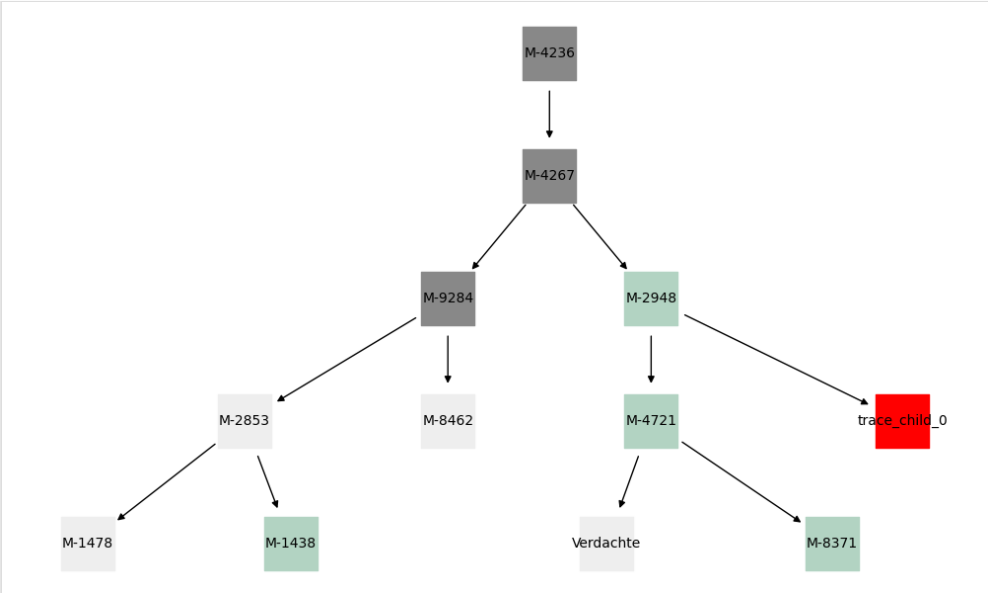
Pedigree Structure

Total Individuals: 11
Known Haplotypes: 4
Root Individual: M-2948


Average inside pedigree probabilities
0.000192
0.000190
0.000188
0.000186
0.000184
0.000182
0.000180
0.000178
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
iterations (x1000)


Average outside pedigree probabilities
0.104
0.102
0.100
0.098
0.096
0.094
probability
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
iterations (x1000)





Marker Set

Number of Markers: 30
Marker Names: DYF1000, DYF1001, DYF1002, DYF387S1, DYF393S1, DYF399S1, DYF403S1a, DYF403S1b, DYF404S1, DYR88 ...

Simulation Parameters

Two-step mutation factor	0.03
Stability window	10000

Model validity threshold	0.0500
Bias value	Adaptive

Model Convergence Plots